

Boxes Counter Automation using PLC

Key Objectives

- Develop a PLC-based automation system that counts boxes on a conveyor and organizes them into batches.
- Trigger box production every 3 seconds and manage buffer/entry conveyors to prevent batch collisions.
- Use sensors to track box positions and batch counts.
- Provide visual feedback via LEDs and Factory IO lamps.
- Implement logic for system stop at the 9th and 15th box, and include emergency and reset controls.

Technical Overview

The system uses PLCnext to control a virtual conveyor environment in Factory IO. A start button (S1) initiates the box production process. Boxes are emitted every 3 seconds and grouped in batches of 3. The Entry and Buffer conveyors manage movement to ensure spacing between batches. Sensor inputs increment box and batch counters, which are displayed in Factory IO. Upon counting the 9th box, conveyors pause for 3 seconds and the emitter halts for 17.5 seconds. The process resumes until the 15th box is produced and removed, after which everything stops. Emergency stop (S3) halts the system without resetting counters, while reset (S2) clears ongoing activity.

Visual Aids

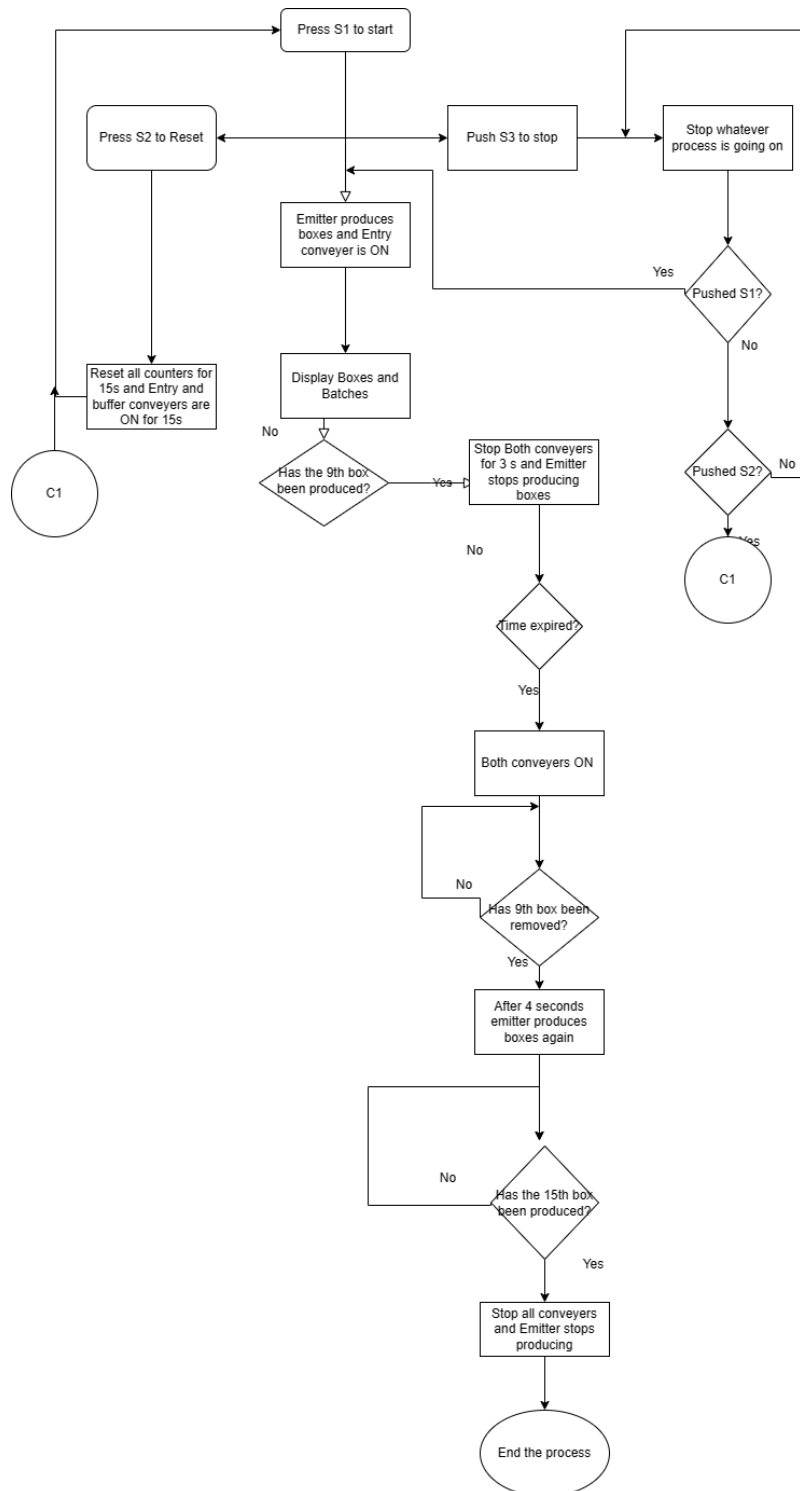
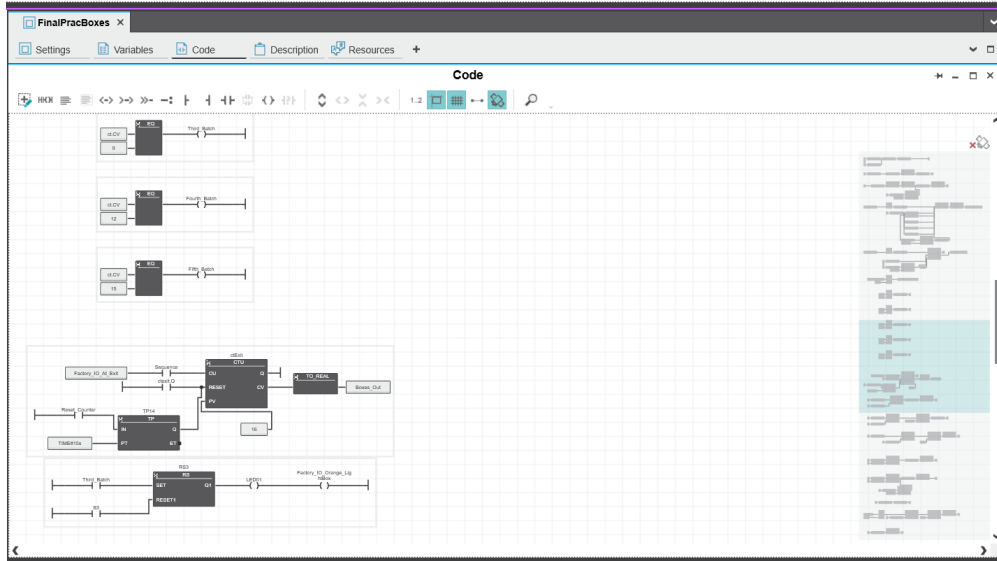


Figure 1: Flowchart diagram for the code



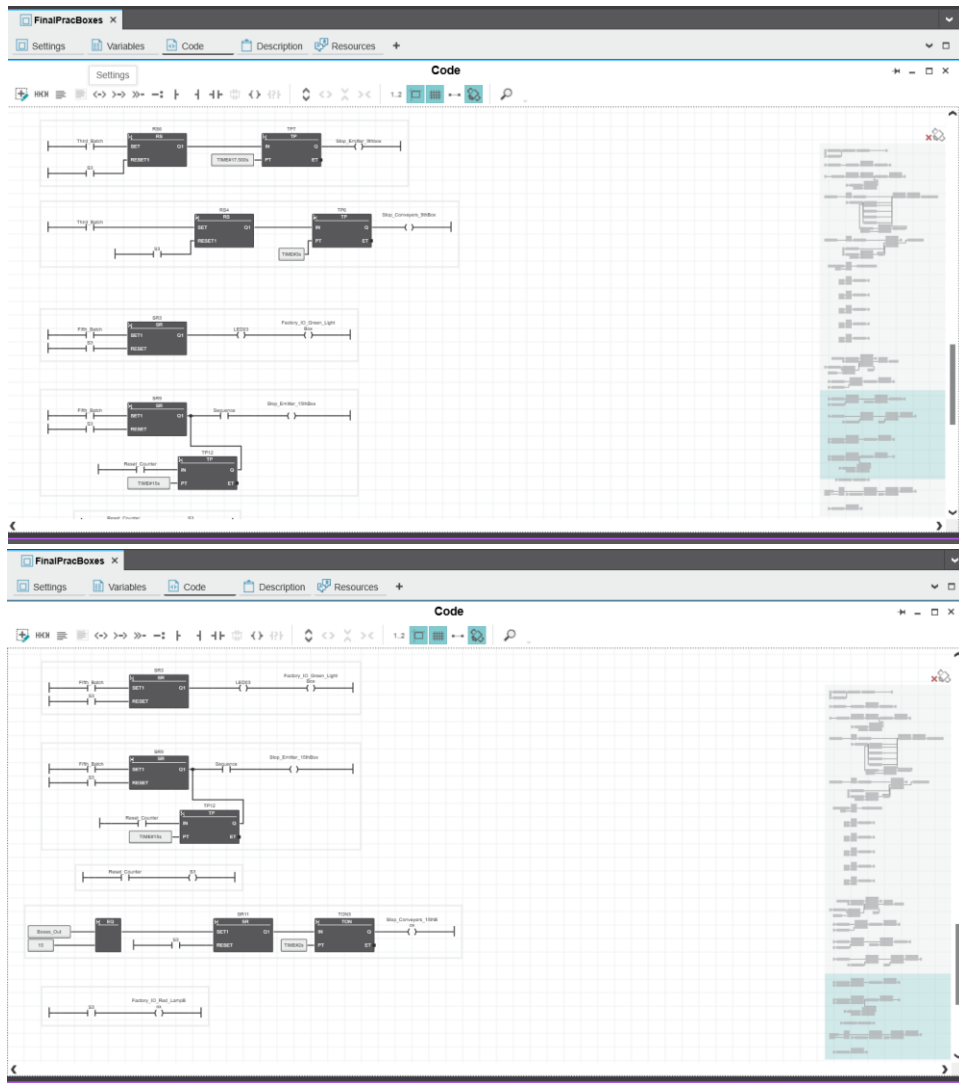


Figure 2: Full ladder logic code divided into five screenshots

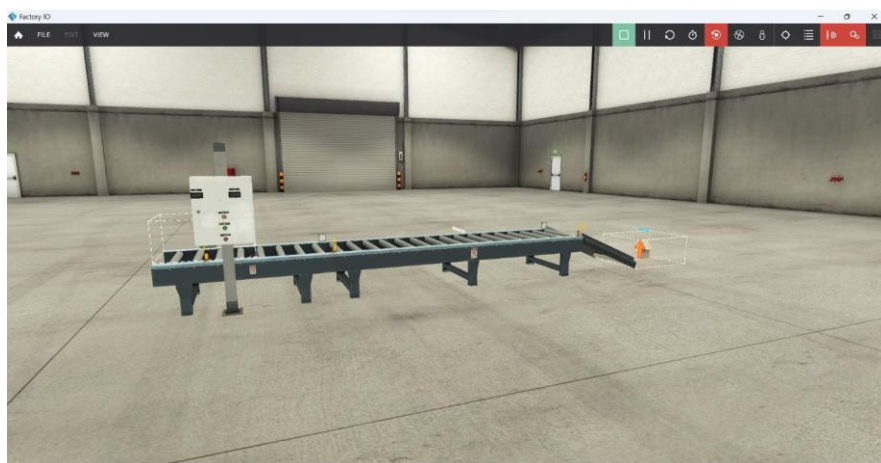


Figure 3: Conveyor belt simulation using Factory IO simulation environment

Highlights

- Effective use of counters, timers, and logical control for real-time simulation.
- Realistic delay and conveyor control to mimic batch grouping and spacing.
- Digital displays for box and batch counts with responsive visual indicators (LEDs and Factory IO lamps).
- Emergency and reset logic allows flexible interruption and resumption without losing count history.
- Batch recognition triggers specific light indicators at milestones (9th and 15th boxes).

Conclusion

The automation system met the design specifications, ensuring boxes were counted and grouped correctly into batches without collision. Reset and stop logic was implemented successfully. One of the challenges was fine-tuning timer delays and ensuring boxes did not overlap between batches. Overall, the system demonstrates solid use of ladder logic principles and integration with Factory IO simulation.